How Scoring Responses Change Attribution Evaluation in a Diffusion Policy

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Abstract

Robots predict many action coordinates, so evaluation scores can disagree. For a diffusion policy, we separate the quantity explained by Integrated Gradients, which uses numerical integration, from the response scoring input removal and restoration. Q is negative squared distance from the reference action; N is negative distance with a small stabilizer. An exact identity shows that, for the same curve, N's normalized area never exceeds Q's when partial input replacement moves predictions no farther than replacing every feature of the tested input type. Larger changes, called overshoots, can reverse this ordering. In saved simulated cube-picking runs of a 170-million-parameter model with no successful episodes, all medians of Q-minus-N differences are positive, but extreme calls make most means negative. For image-patch deletion in three matched groups (130 episodes), N-target Integrated Gradients exceeds Q-target under both responses in pretrained PickCube and StackCube; both beat random under N in all three, but under Q only in StackCube. The raw-mean winner is the method maximizing mean advantage over random ordering. Switching Q and N changes it in 80% of pretrained PickCube deletion resamples, whereas scaled winners match in 90% and 98% for pretrained and task-trained PickCube. Only 1 of 12 image-patch comparisons statistically separates the top two. Checks on twelve matched contexts of one pretrained model show 16-bit versus 32-bit calculations and integration rule affect attribution recovery. These samples demonstrate possible numerical failures, not their frequency, and do not verify the original rankings; response choice, ranking quality and numerical accuracy therefore require separate evidence.

Keywords: integrated gradients, attribution evaluation, diffusion policies, perturbation metrics, numerical integration, robotics, explainable AI

Introduction

Explaining and evaluating robot actions

Attribution methods rank input features, here image patches, language tokens or robot-state coordinates. Testing these explanations can help investigate what influences robot predictions. Insertion starts from a baseline, such as a gray image, and restores highly ranked features first; deletion removes them from the actual input. Fast recovery during insertion and fast change during deletion conventionally favor an explanation[1]. Both tests need a response that reduces prediction change to one score.

Unlike a classifier with a class score, this robot predicts 512 action coordinates per call. Evaluators must choose which output quantity an explanation concerns[2]. Huber and colleagues found different preferred methods for action preference and state value, and instability in endpoint normalization[3]. Zhang and colleagues score vision-language-action interventions with action mean squared error; interpreting it through a Gaussian policy requires assumptions that need not hold for a multimodal diffusion policy[4]. Response choice therefore needs explicit testing.

Diffusion policies refine random noise into action sequences[5]. The Robotics Diffusion Transformer (RDT) conditions predictions on images, instructions and state[6]. Integrated Gradients (IG)

multiplies each feature's input change by its average gradient along the baseline-to-input path[7]. The *target* is the quantity explained; the *response* scores predictions after input removal or restoration. For example, Q-target IG explains negative squared action distance while N scores the same predictions by negative stabilized distance. Common practice uses one scalar for both; separating them changes the score without changing rankings or predictions.

What existing evaluation methods establish

Removal-based explanations vary the intervention, model behavior and summary; metrics can follow this structure[8]. Retraining[9], information conveyed by which inputs are hidden[10], and implementation choices[11] affect what scores measure. Insertion/deletion areas under the curve (AUCs) retain feature order but discard attribution magnitude[12]. Meta-evaluation tests metric reliability[13]; rankings need not transfer between datasets[14]. Time-series tests show that low-ranked features can reveal information missed by testing only high-ranked ones[15]. Evaluation thus requires controls and explicit scope.

Input replacement can produce examples outside the training distribution[16]. Ways of hiding inputs change rankings[17]; single-deletion benchmarks address distribution shift and model comparisons[18]. F-Fidelity fine-tunes a replacement model by hiding inputs independently of the explanation and caps evaluation removals at the same budget[19]. Diffusion reconstruction changes masked images[20], while missingness-bias calibration corrects classifier outputs[21]. Here interventions stay fixed: only the scoring response changes.

Output-dependent preferences are already known: TRACE finds different winners for probabilities and unnormalized scores, or logits[22]. Hama, Mase and Owen analyze interactions, response transformations and random-order references[23]; regression baselines also change the question asked[24]. Counterfactual tests challenge visual attribution interpretations[25], StateMask identifies reward-relevant time steps[26], and internal-activation changes can steer vision-language-action behavior[27]. These different behavioral questions motivate specifying the target and response. This study adds an endpoint-normalized identity and tests matched robot contexts.

Completeness means that attributions sum to the baseline-to-input target change, assuming path gradients integrate to that change[28], but does not establish usefulness[29]. Alternative paths change the explanation[30]; IDGI redirects sampled contributions rather than improving quadrature for standard IG[31]. Ground-truth benchmarks[32], synthetic simulations[33] and model-randomization critiques[34] test other properties. Signed contributions may cancel in metrics[35]. Integration-budget checks[36] complement ranking evaluation, but do not establish accuracy for this diffusion policy in 16-bit arithmetic.

We examine response changes to fixed curves, their appearance in successful and failed episodes, criterion-dependent method preferences, and precision and integration-rule effects on recovery, in that order. For saved Q-ranked vision deletion, median AUC is 0.448 under Q versus 0.291 under N: an illustrative, unvalidated 0.40 cutoff judges identical predictions differently.

Methods

Experiments and preserved evidence

An *episode* is one simulated task attempt; a *call* predicts 64 timesteps of seven joint values and one gripper value. A *context* fixes that call's observation, instruction, state and diffusion noise. Two saved PickCube runs each contain thirty episodes and 750 calls, all unsuccessful. PickCube requires moving a cube to a goal. One run ranked by Q, the other by N; both used pretrained RDT-

170M without task fine-tuning[37]. *Rescoring* computes another response from saved intervention predictions, without rerunning the robot or recovering missing observations and noise.

Matched groups are pretrained PickCube (60 episodes, 120 contexts), task-trained PickCube (40, 80), and pretrained StackCube (30, 60): respectively, the saved setting, frequent success, and cube stacking. The RDT developers' task-trained RDT-1B has about 1.2 billion parameters versus 170 million. Per episode, two calls used by the simulator are selected uniformly without replacement under a rule fixed before collection. Numerical tests add all 21 available contexts from the collection underlying twelve completed offline diagnostics, totaling 281 contexts. Separate saved runs cover task-trained PickCube, side peg insertion and household-object picking.

Identities are recorded, recovered from unchanged file histories, inferred, or unknown (Supplementary Table S2). Saved code requested the latest model versions rather than recording a revision. Both saved runs' weights are recovered as the same files fixed for the matched study: revision `8aa386cac3bb`, `pytorch_model.bin`, hash prefix `2e87fccee029`; no run-time hash exists. SigLIP image-encoder revision `9fdffc58afc9` and preprocessing (384-by-384 bicubic, scale 1/255, mean/standard deviation 0.5) are recovered; original software versions are mostly unknown. Five denoising steps with the DPM-Solver++ sampler[38] are recorded for Q and recovered for N. Resetting the seed at each call reused one noise draw per episode; missing inputs still prevent exact regeneration.

The matched study preserves model/code versions, observations, state, noise, predictions, rankings, masks and settings. Three machines recorded Python 3.12.3 and Torch 2.8.0+cu128; the fourth runtime is inferred. A fresh machine reproduced 596 attribution maps and 2,466 AUC values for one context bit for bit; two further contexts matched. This verifies those reconstructions, not general hardware independence.

Targets and input replacement

The simulation runs in the ManiSkill3 environment[39]. Our controller executes every fourth predicted action, at most 16 simulator steps per call, and stops an episode by 400 steps. An input *modality* is vision, language or robot state. Holding all other inputs and the noise fixed, let $g(u)$ be the 512-dimensional action prediction when one modality is u. The actual input is x, its chosen baseline is x', and the reference action is $a = g(x)$. Define

$$r(u) = \|a - g(u)\|_2^2, \quad Q(u) = -\frac{r(u)}{2d_a}, \quad N(u) = -\sqrt{r(u) + \epsilon}, \quad d_a = 512, \quad \epsilon = 10^{-12}. \quad (1)$$

Here r is squared action distance. Q divides its negative by twice the action-coordinate count, resembling a Gaussian log-likelihood without representing the policy's likelihood. N uses stabilized distance to keep its gradient finite at zero. The reference action and noise stay fixed throughout each curve.

Vision features are 729 patches in a 27-by-27 external-image grid; five other camera slots stay fixed. A gray-image representation supplies the baseline. Released SigLIP and T5 variants encode images and language, respectively[40,41]. Language features are attended token positions after the adaptor. Saved Q rankings use encoded padding/end-of-sequence tokens; saved N rankings use the RDT developers' precomputed representation with a zeroed baseline. These baselines prevent cross-run language ranking comparisons. State has seven joint readings and one gripper reading, baseline zero. Baselines are not uniquely neutral[42]; interpolated representations may not encode valid language[43].

Let $B = r(x') > 0$ measure the squared action change when every feature of the tested modality is replaced. For either response $S \in \{Q, N\}$,

$$C_S(u) = \frac{S(u) - S(x')}{S(x) - S(x')}, \qquad \Delta = \mathrm{AUC}_Q - \mathrm{AUC}_N. \tag{2}$$

The normalized response is zero at the tested modality's baseline and one at its actual input; other modalities stay fixed. AUC integrates it over replacement fractions with identical trapezoidal weights for Q and N. A *paired* difference scores one curve under both responses. An *overshoot* means partial replacement moves predictions farther than replacing every feature of that modality, $r > B$. Responses and AUCs can leave the range zero to one and be very negative when B is small. Normalization is undefined at $B = 0$.

Saved curves use nominal percentages 0, 1, 5, 10, 20, 30, 50, 75 and 100. At an intermediate percentage p, $k = \max(1, \mathrm{round}(Gp/100))$ groups are selected, with half-integer ties rounded to the nearest even integer. Here G counts attended language-token positions. Reconstructed tokenization gives 20, 30, 25 and 16 for PickCube, StackCube, PegInsertionSide and PickSingleYCB. Missing original masks limit run-specific recovery. For $G = 16, 20, 25$, both 1% and 5% select one token: all 3,000 saved PickCube language curves (two runs, 750 calls and two directions) have equal responses at those points. The 800 saved StackCube curves do not. Their composition and realized-fraction checks appear in the supplement. Saved rescoring retains nominal fractions; matched experiments use preserved masks and actual selected fractions.

Integration and matched method comparisons

For target $F \in \{Q, N\}$, IG multiplies each feature's input difference by its average path gradient. With m integration intervals,

$$A_i^F = (x_i - x_i') \int_0^1 f_i(\alpha)\, d\alpha, \qquad E_{i,m} = \frac{x_i - x_i'}{m+1} \sum_{j=0}^{m} f_i(j/m), \tag{3}$$

where $f_i(\alpha) = \partial_i F(x' + \alpha(x - x'))$. The equal-weight estimator E gives the same weight to all $m + 1$ points. The trapezoidal estimate T halves the endpoint weights[44], so exactly

$$E_{i,m} = \frac{mT_{i,m} + (x_i - x_i')[f_i(0) + f_i(1)]/2}{m+1}. \tag{4}$$

The extra endpoint weight can produce error proportional to $1/m$. Exact integration in piecewise-linear networks also exposes sampling error[45]. The saved estimator used 16-bit bfloat16 arithmetic, approximately three significant digits; its 64-interval budget is inferred for the Q-ranked run. It replaced NaN or infinite gradients with zero without counting them. The numerical accuracy of the saved rankings is unverified.

Numerical variants change model precision, summation and rule (Table 1; Supplementary Table S8). Corrected 16-bit calculations sum at 32 bits, about seven significant digits. Midpoint and Gauss-Legendre rules use interior nodes; adaptive Gauss-Kronrod adds nodes using an error estimate. Without an exact policy answer, qualification requires adaptive convergence and pairwise agreement among adaptive integration, 128-node Gauss-Legendre and the 512-interval 32-bit trapezoid: relative L1 difference below 0.1%. Relative L1 is the absolute coordinate-difference sum divided by the absolute comparator-coordinate sum. Supplementary Table S5 separates the diagnostic populations.

Table 1 | Study groups answer different questions. Counts are episodes (successes) / calls unless stated. Precision is model / accumulation: bf16 is 16-bit, fp32 32-bit. Budgets count intervals; equal weighting uses one extra evaluation. Asterisks indicate inferred budgets. The saved 1B rerun is excluded. Preserved and offline samples overlap. Saved runs measure response shifts, matched groups compare methods, and other contexts test numerics. Saved 170M weights are recovered; matched weights are recorded. Tables S2 and S8 give full identities, statuses and variants.

Group	Episodes / calls	Rule and budget	Precision	Preserved
Saved PickCube 170M, target Q	30 (0) / 750	Equal weights, 64*	bf16 / bf16	Scored curves
Saved PickCube 170M, target N	30 (0) / 750	Equal weights, 64	bf16 / bf16	Scored curves
Saved PickCube 1B, target Q	60 (15) / 1,199	Equal weights, 64	bf16 / bf16	Scored curves
Saved peg and object tasks, 170M, target Q	16 (0) / 400 each	Equal weights, 128*	bf16 / bf16	Scored curves
New PickCube 170M, targets Q/N	60 (0) / 120	Trapezoid, 512; checks	bf16 / fp32; fp32 subset	Complete contexts
New PickCube 1B, targets Q/N	40 (31) / 80	Trapezoid, 512; old rule	bf16 / fp32	Complete contexts
New StackCube 170M, targets Q/N	30 (0) / 60	Trapezoid, 512; old rule	bf16 / fp32	Complete contexts
Preserved four-task sample	21 contexts	Numerical checks	bf16 and fp32	Contexts
Separate offline diagnostics	6 / 12 per input and target	Trapezoid, 128 to 1,024	fp32 / fp32	Contexts and outputs

Even a linear action example exposes endpoint sensitivity. For $g(u) = u$, baseline zero and input one, exact N attribution is 0.999999, slightly below one because of the stabilizer. At 64 intervals, equal endpoint weights underestimate it by 1.54%, and trapezoidal weights by 0.78%. Midpoint and 64-node Gauss-Legendre estimates slightly exceed it, each by about one part per million. N's gradient falls sharply to zero near the actual-input endpoint. Over the tested budgets, that unresolved transition gives approximately first-order trapezoidal error. With fixed positive epsilon the integrand is smooth and this is not a universal asymptotic claim.

Seven methods rank groups by summed absolute values: Q/N IG, Q/N mean path gradients, input difference, and Q/N original equal-weight IG. Path gradients omit IG's input-difference factor; input difference ranks baseline-to-input change alone. Ordinary IG/path gradients use 512 trapezoidal intervals, a 16-bit model and 32-bit accumulation. Chance controls average 32 image orders, 64 token orders and all $8! = 40{,}320$ state orders. Raster, reverse and center-out image orders, plus forward/reverse token or joint orders, test attribution-free alternatives.

Advantage is method-minus-random AUC for insertion and random-minus-method for deletion; larger values favor a method. The primary criterion averages within episodes, then across episodes with defined values for every method. A common random reference cannot reorder methods within one response. Scaling context advantages by their random-order standard deviations before episode averaging changes weights and may change winners; the state reference enumerates every order. Episode medians and illustrative deletion cutoffs of 0.40 and 0.50 check sensitivity.

The collection protocol, analysis plan and three amendments were recorded before their governed outcomes were examined. The seven primary pretrained-PickCube vision-deletion contrasts in Table 2A use 10,000 bootstrap draws that resample entire episodes to retain dependence between their calls. Bonferroni adjustment divides the error allowance among seven comparisons, giving alpha $0.05/7$. Best-versus-second intervals use $0.05/4$ for each group's four cells, two responses times two directions. Equal highest values have no unique winner. These selected comparisons are exploratory and do not jointly test a winner across all groups. Simulated interval coverage was checked before examining results. Saved-run medians and positive shares were emphasized retrospectively because means are tail-sensitive, but all means remain reported. Intervals describe recorded episodes, not variation across training runs, checkpoints or tasks.

A separate retrospective report uses 10,000 paired episode-bootstrap draws, seed 930011, to count matching winners, different winners and ties under raw means, scaled means and episode medians. Success/failure median intervals resample whole episodes within each outcome. These describe observed-sample stability, not probabilities of a true population reversal.

Completeness residual is the absolute attribution-sum error divided by the absolute target change. Coordinate differences, Spearman rank correlation, the share of top-ranked groups in common, overlap of replaced-feature sets and AUC changes are reported separately. Zero denominators remain undefined. The 1% completeness and 0.01 AUC-change thresholds are study-specific diagnostics, not guarantees. Separate 32-bit diagnostics cover twelve contexts per modality/target from six episodes across four tasks, with no completed RDT-1B context. They demonstrate possible numerical failures without estimating frequency.

All experiments are simulated, involve no human or animal subjects and require no IRB approval.

Figure 1 connects the workflow, scoring responses and constructed examples.

Results

Response identity and saved curves

Exact self-reference means that the reference action is the model's own actual-input prediction, giving $r(x) = 0$. The saved curves satisfy this to floating-point precision. Equation 2 yields

$$C_Q - C_N = \frac{(\sqrt{r+\epsilon} - \sqrt{\epsilon})(\sqrt{B+\epsilon} - \sqrt{r+\epsilon})}{B}. \tag{5}$$

The difference is positive when $0 < r < B$, zero at $r = 0, B$, and negative at overshoots. If every sampled point is in range, $r \leq B$, then $\Delta \geq 0$, strictly when some positive-weight point satisfies $0 < r < B$. Thus N makes the same ranking look better for deletion and worse for insertion. The saved schedule gives $\Delta \leq 0.2175$ for every defined curve. Figure 1B–D illustrates the score, identity and a constructed target reversal; proofs are in Supplementary Equations S1–S6.

The usual 0.5 benchmark corresponds to a straight-line normalized response; random-order areas need not equal it. If every feature contributes equally and independently, every ordering gives

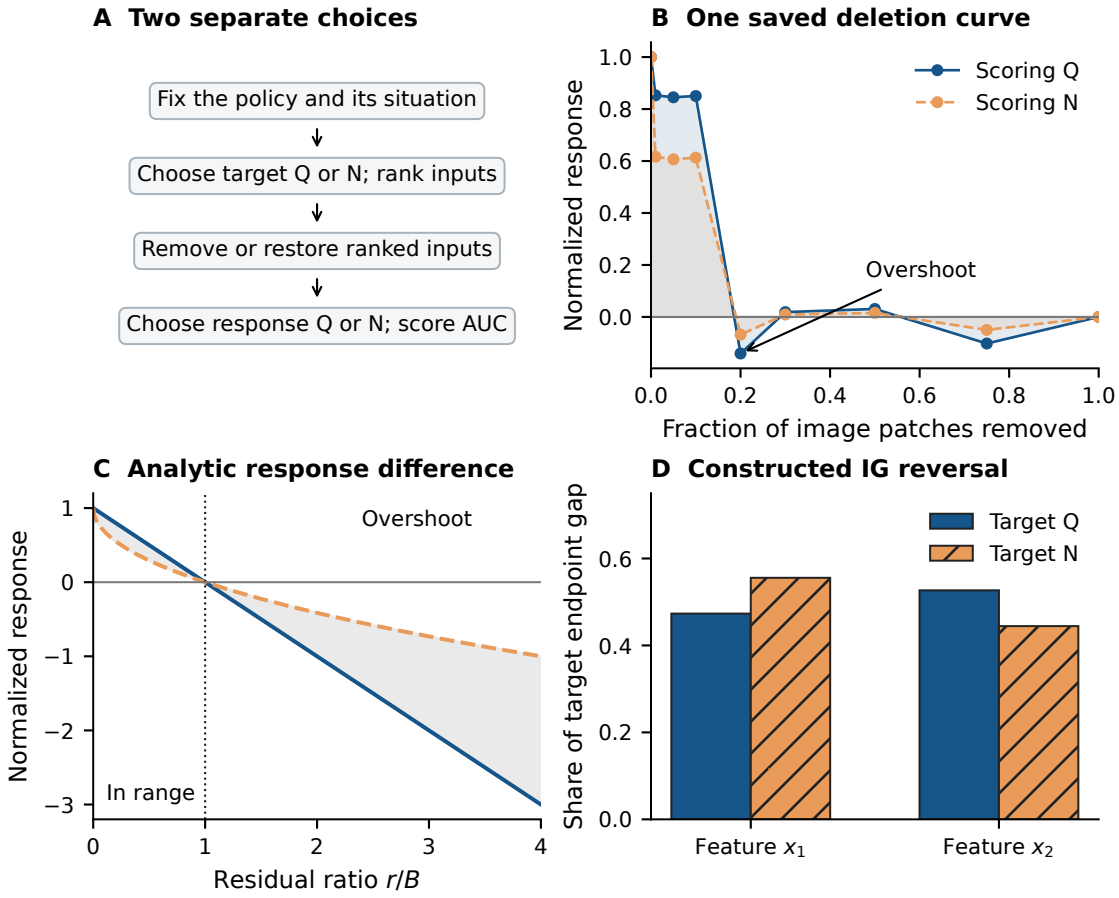


Figure 1 | Target and scoring response are separate choices. A) Workflow. **B)** One actual saved curve scored by Q and N; shading defines AUC and $r > B$ marks an overshoot. **C)** Analytic responses versus squared residual ratio r/B, with their gap in the zero-stabilizer limit. **D)** IG shares for $g(x_1, x_2) = x_1^2 + 0.8x_2$, divided by each target's total change. Labels identify input features. The constructed reversal is not an observed PickCube result; the supplement gives the function and proof.

insertion/deletion AUC 0.65976 under Q on the saved 100-feature schedule, versus 0.5 under unstabilized N. Thus 0.5 misstates Q's chance performance. These are analytic examples, not RDT controls. A three-feature construction also reverses ordering preference by changing only the response (supplement).

All eight saved-case medians are positive: two ranking targets, two modalities and two directions. Medians range from 0.1041 to 0.1590; 75.6%–94.3% of calls have $\Delta > 0$, and 16.0%–74.0% overshoot. These medians measure the size of a response shift whose fully in-range direction is predicted; positivity of a median in an overshooting population remains empirical. Six means are negative, and three corresponding intervals span zero. For Q-ranked vision insertion, the mean is -17.1: the 30 first calls contribute -16.92 and the remaining 720 contribute -0.18, both divided by all 750 calls. The same seven of thirty episodes supply differences below -100 in both saved runs, so their tails are not independent confirmations.

Figure 2B shows 29–144 of 750 calls per case below -0.05, none above 0.25: positive centers hide substantial tails. Supplementary Table S1 gives full summaries. Shared simulator seeds do not

match contexts: Pearson correlation of log B across thirty aligned episodes falls from 0.992 at call 0 to 0.289 at call 2, then ranges around 0.4–0.8 and ends at 0.66. Head-to-head ranking comparisons are therefore invalid. N-ranked vision B has fifth/ninety-fifth percentiles 2.9/80.8.

Small B magnifies intermediate prediction changes. Across 10,796 saved case calls, 335 have $\Delta < -1$ and 23 have $\Delta < -100$. Equation 5 itself requires the maximum interior residual ratio to exceed approximately 2.83 and 126.2, respectively; these thresholds are not independent evidence for a mechanism. Observed minimum ratios are 3.83 and 252.8, and the most extreme calls have $B \leq 0.514$. Concentration in first calls and correlations from -0.56 to -0.98 across sixteen overshooting groups support a possible small-denominator mechanism. Whether partially replaced inputs overshoot because they resemble nothing in training remains untested. Supplementary Figure S1 shows the detailed geometry.

Successful episodes and additional tasks

Saved task-trained PickCube has fifteen successful episodes (74 calls) and forty-five failed episodes (1,125 calls) across three seeds. Successes stop early, emphasizing early calls. Successful/failed vision medians are 0.0653/0.0868 for insertion and 0.0747/0.1035 for deletion. Overshoot rates are 24.3%/29.0% and 41.9%/31.6%, respectively. More extreme failure minima partly reflect fifteen times as many calls. Supplementary Table S6 gives rates and intervals, keeping language separate by seed because baselines differ.

Task-trained PickCube succeeds in 31/40 episodes (62 successful, 18 failed calls); both pretrained groups have no successes. Q-target trapezoidal IG gives successful/failed medians 0.0616/0.0913 for insertion and 0.0212/-0.0086 for deletion. The failed deletion interval $[-0.1775, 0.0608]$ rests on nine episodes; its sign is not interpreted. Deletion overshoots affect 41/62 successful and 13/18 failed calls; insertion counts are 13/62 and 7/18. The largest extremes occur in successes, opposite the saved pattern, but successes have more observations. Different estimators, contexts and runtimes preclude causal success claims.

The saved peg-insertion and household-object tasks each contain sixteen unsuccessful episodes and 400 calls, using a 128-interval estimator. Their vision insertion/deletion medians are 0.0990/0.1152 and 0.0954/0.1405. Peg insertion retains a large negative insertion tail: mean -3.54, minimum about -212, and overshoot share 69.0%. Figure 2 compares these separate populations. All successful evidence in this study comes from PickCube.

Matched rankings and criterion sensitivity

For image-patch deletion on sixty pretrained-PickCube episodes, N-target IG exceeds Q-target IG under both responses (Table 2A). Both exceed random ordering under N; neither separates from random under Q, whose tails widen intervals. Changing both target and response gives -0.438, adjusted interval $[-3.91, 2.44]$, an inconclusive comparison in different normalized units.

Descriptive 95% percentile deletion intervals favor N-target over Q-target IG under both responses in StackCube, but not task-trained PickCube. Both variants exceed random under N in all three groups; under Q, only in StackCube. The task-trained N-target-versus-random comparison under N includes zero under the bias-corrected and accelerated (BCa) interval (Supplementary Table S3).

Image-patch deletion point estimates give different raw-mean winners under Q/N in all three groups; scaled and episode-median winners differ only in StackCube (Table 2B). Raw winners differ in 79.63% of pretrained-PickCube resamples versus 30.05% in StackCube. Scaled winners

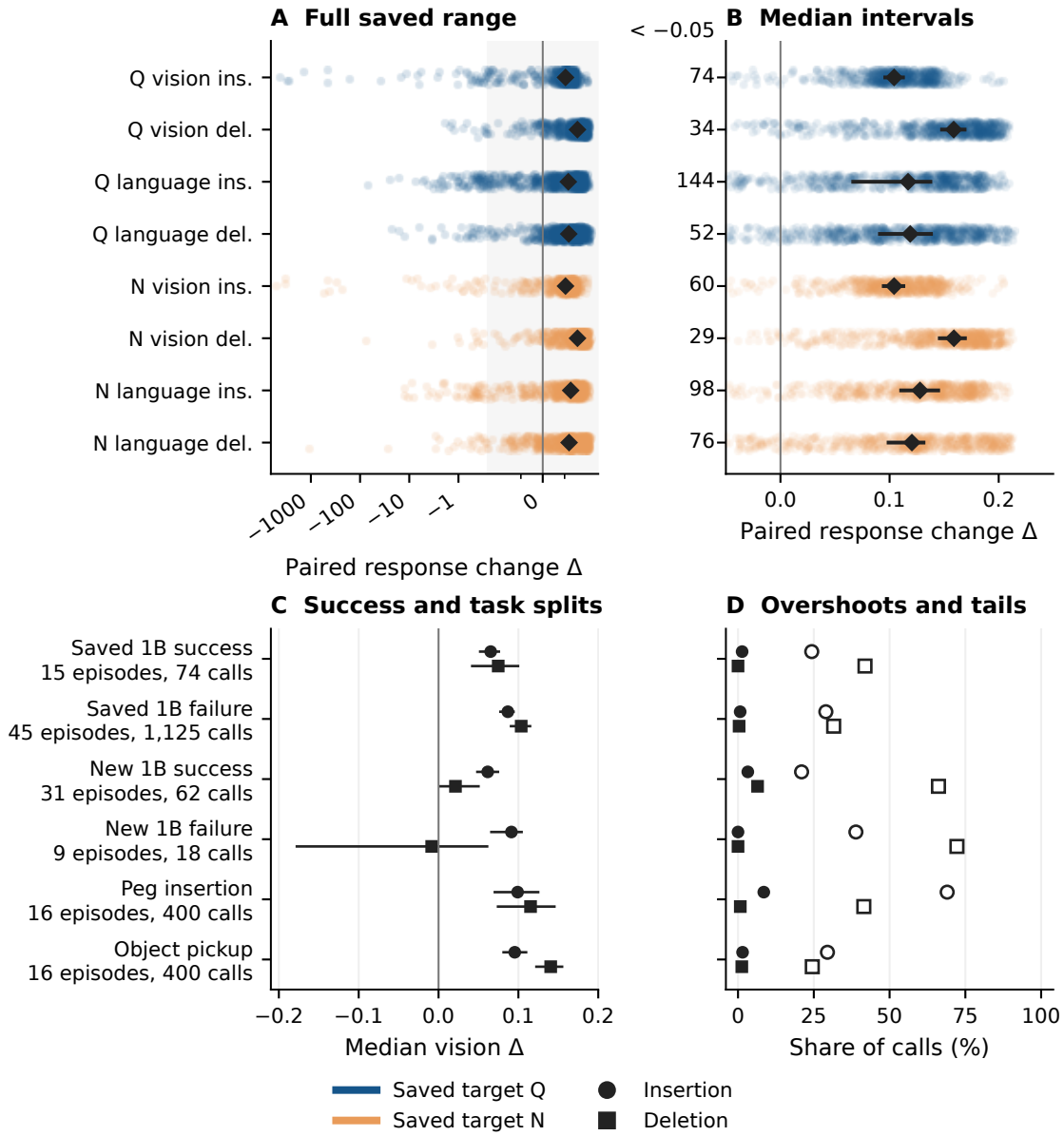


Figure 2 | Positive centers coexist with tails in successful and failed episodes. A) All eight saved distributions, 750 calls each. Symmetric-log scaling is linear for $|\Delta| \leq 0.25$, compressing larger magnitudes without losing sign. **B)** Central medians, 95% episode-bootstrap intervals and left-edge counts below -0.05. Q/N row labels denote ranking targets; $\Delta = \mathrm{AUC}_Q - \mathrm{AUC}_N$. **C)** Vision medians and intervals for separate success/failure and additional-task populations. **D)** Overshoot shares (open) and $\Delta < -1$ shares (filled); circles: insertion, squares: deletion. Table 1 identifies estimators; Supplementary Tables S1 and S6 give full values and seed membership. Repeated calls are dependent.

Table 2 | Primary contrasts and response-dependent choices. A: seven preplanned image-patch deletion contrasts, 60 pretrained-PickCube episodes, Bonferroni-adjusted intervals. The last row mixes response units. B: image-patch deletion point-estimate winners. Scaling divides each call's advantage by its random-order standard deviation before averaging; medians use episode means. IG uses 512 trapezoidal intervals; equal-weight IG uses 64 intervals. Both use 16-bit models. Winners need not separate statistically. Supplementary Tables S3 and S4 give all comparisons and controls.

A. Primary contrasts

Contrast	Q: estimate [interval]	N: estimate [interval]
IG-Q minus random	0.0280 [−3.54, 3.12]	0.419 [0.327, 0.556]
IG-N minus random	2.89 [−3.41, 14.6]	0.466 [0.341, 0.704]
IG-Q minus IG-N	−2.86 [−11.4, −0.0115]	−0.0467 [−0.146, −0.00846]
IG-Q at Q minus IG-N at N	−0.438 [−3.91, 2.44]	Mixed responses

B. Point-estimate choices

Group (episodes)	Score	Raw mean	Scaled mean	Episode median
PickCube 170M (60)	Q	Path-gradient N	IG-N	IG-N
	N	Equal-weight IG-Q	IG-N	IG-N
PickCube 1B (40)	Q	Input difference	Path-gradient N	Path-gradient N
	N	Path-gradient N	Path-gradient N	Path-gradient N
StackCube 170M (30)	Q	Path-gradient N	Path-gradient N	Path-gradient N
	N	Equal-weight IG-N	Equal-weight IG-N	IG-N

match in 90.02% and 98.48% for pretrained and task-trained PickCube; median agreement is less stable (Supplementary Table S7). Only one of twelve image-patch top-two intervals excludes zero: pretrained PickCube Q deletion, 0.0998, adjusted interval $[0.0181, 0.315]$. Two language cells also separate, three of twenty overall (Supplementary Table S3).

Fixed orders beat every attribution method's raw mean in three vision cells: pretrained PickCube insertion under Q and N, and task-trained PickCube deletion under Q. The latter winner, input difference, also falls below random. Scaling and medians change these comparisons (Supplementary Table S4). Unvalidated 0.40 and 0.50 cutoffs each disagree with calibrated preferences in four of six deletion cells. Supplementary Table S7 reports bootstrap winner stability with ties separate.

Numerical recovery and diagnostic limits

Figure 3 compares matched numerical recovery.

Most saved vision rankings miss the study's 1% completeness criterion: median residuals are 1.96% for Q-ranked and 4.76% for N-ranked calls, with 77.2% and 99.6% above 1%. Their sums fall short of the baseline-to-input target change in 87.5% and 99.1%. This one-sided deficit fits endpoint bias; storage rounding alone can err in either direction. Individual coordinate errors remain unknown.

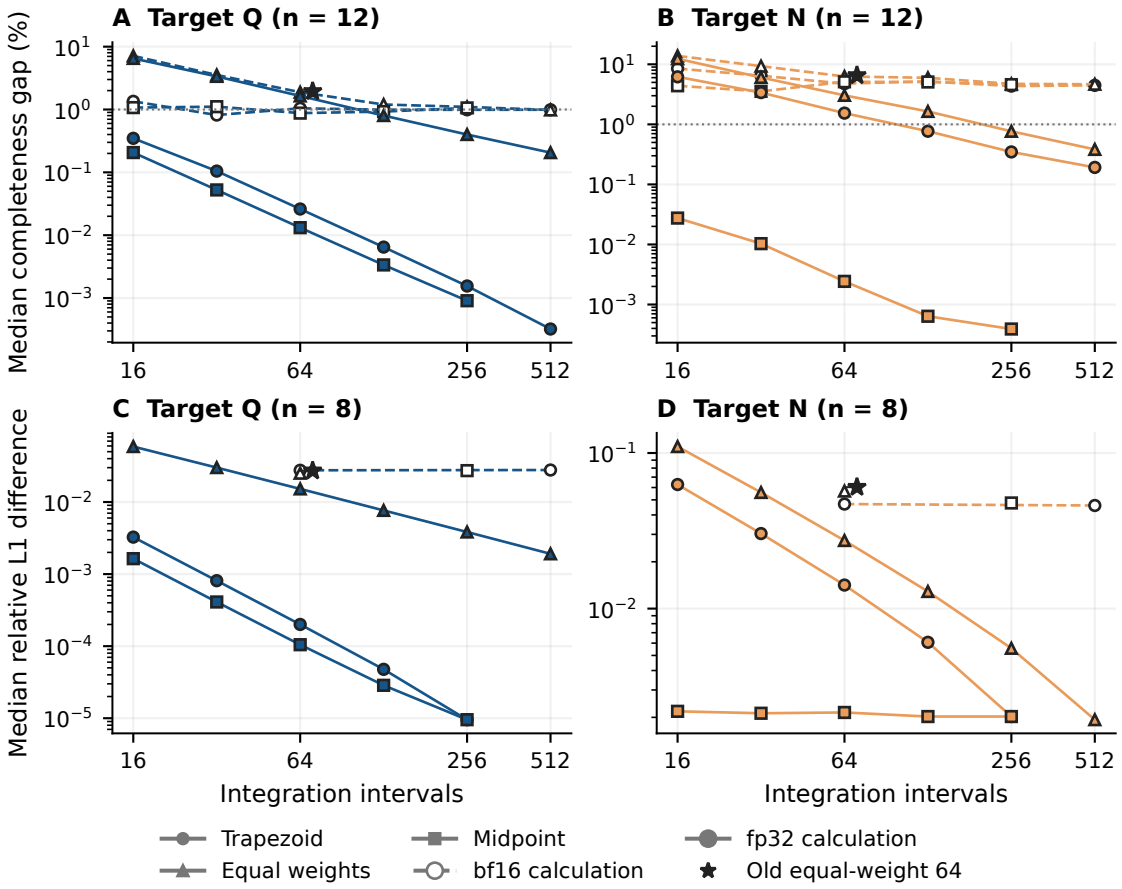


Figure 3 | Precision and rule affect recovery on matched contexts. A)–B) Median completeness residuals, twelve pretrained-PickCube vision contexts; horizontal line: study-specific 1% threshold. **C)–D)** Relative L1 differences from the 32-bit, 512-interval trapezoid, eight stored-map contexts. Open/dashed markers/lines indicate 16-bit calculations, filled/solid 32-bit; shapes identify rules. Original-estimator stars are offset for visibility. The comparator is not exact IG: its N endpoint discrepancy contributes to D's midpoint plateau. Its zero self-difference is omitted on logarithmic axes.

On twelve matched pretrained-PickCube vision contexts, increasing trapezoidal intervals from 64 to 512 changes 16-bit median completeness residuals from 4.76% to 4.46% for N and 1.05% to 0.991% for Q. With 32-bit model calculations, N falls from 1.54% to 0.192% and Q from 0.0261% to 0.000321%. Additional 16-bit points help little. At sixteen intervals, 32-bit midpoint N reaches about 0.0275%. Rule and interval count both matter.

On 120 new pretrained-PickCube vision contexts, original-estimator median relative L1 differences from the corrected 16-bit trapezoid are 2.07% for N and 1.82% for Q. The 32-bit subset has twelve contexts; only eight have stored 16-bit rule-comparison maps (Supplementary Table S5). Rank correlations stay at least 0.9993; median top-5% overlap is one. Yet N-target deletion AUC changes exceed 0.01 in 7/120 N-scored and 12/120 Q-scored contexts; Q-target counts are 4/120 and 12/120. The latter median/max are 0.0018/221: normalized AUC is unbounded below, so stable rankings need not give stable scores.

All 815,210 evaluated nodes are finite; no 64-bit model run was performed. Adaptive integration converges in all 48 comparisons, but three-way qualification passes only 15/24 Q and 5/24 N cases: eight contexts (four new, four stored), three modalities and two targets. For N vision/language, adaptive and Gauss-Legendre estimates agree while the trapezoid differs by about 0.17%; state also differs between the first two. Endpoint effects are plausible, but qualification remains failed and agreement does not prove accuracy.

The separate four-task sample shows agreement of repeated function evaluations despite attribution changes with integration budget (Supplementary Figure S4; Table S5). It demonstrates possible numerical failures, not their frequency or the saved rankings' correctness. The supplement's smooth constructed example passes completeness and grid-agreement checks while reversing exact coordinate order. This is not an observed policy-ranking reversal.

Discussion

Both successful and failed runs have overshoot tails. Primary matched contrasts favor N-target IG, while exploratory winners depend on averaging and scaling: eleven of twelve image-patch top-two intervals include zero. Raw-mean preference does not establish general superiority.

The findings extend Huber's and TRACE's output-dependent preferences. Hama considers monotone transformations[23]; N is strictly increasing in Q. Its condition is a strictly increasing transformation of additive feature contributions on binary inputs. That can hold here since $x_1^2 = x_1$ for binary inputs. Its IG argument uses constant feature slopes before transformation, whereas our quadratic construction changes gradient direction. This different setting is not a counterexample. Our additions concern normalization, overshoots, separate targets/responses and matched numerical comparisons. These findings extend Makino's integration-budget tests to precision and rule choice. Zhu reconstructs interventions; here predictions stay fixed.

Under Q, percentile IG-versus-random deletion intervals include zero in both PickCube groups but not StackCube. Nine failed episodes give an uncertain median; successes show larger extremes. Neither implies causation. Numerical qualification fails despite plausible endpoint effects; most saved rankings miss the completeness criterion. The 0.448/0.291 example shows sensitivity to the illustrative 0.40 rule, not justification for it.

Report targets and responses because they answer different questions; preserve references, noise, baselines and groups for replication. Random and fixed controls reveal what scores reward. Means, medians, tails and undefined cases describe different sample properties; episode resampling retains call dependence. Check completeness, coordinates, ranks and scores separately: none guarantees the others.

Limitations and conclusion

The saved rankings' numerical accuracy remains unverified; missing inputs prevent regeneration and their Q/N runs are unmatched. Original pretrained PickCube evidence is entirely unsuccessful; all successful evidence comes from task-trained PickCube. Only one model family, simulation, two responses, chosen baselines/groups and mostly gradient-based methods are studied. No weight-randomization test of dependence on learned weights was performed[46]. The tail mechanism, including distribution shift from partial replacement, remains untested. Small numerical samples show possible failures, not frequencies. Other successful policies, responses, attribution families and physical robots require matched studies.

The identity proves in-range ordering for fixed predictions. Unsuccessful simulated PickCube runs show its size and tail-dominated means; successful runs also overshoot. Primary comparisons favor N-target IG. Image-patch deletion point estimates change raw-mean winners in all three groups, but scaled and median winners only in StackCube; bootstrap stability varies. Numerical experiments show possible failures, not their frequency, and do not validate original rankings. Scoring choice, numerical recovery and ranking quality require separate evidence.

Data and code availability

A CPU reproduces saved response summaries, plots, success splits and completeness checks. Missing observations and noise prevent original execution and ranking regeneration. Matched inputs support verified fresh-machine reconstruction. Packages supply code, plans, records, identities, supplementary results and replay instructions. The exact source and saved analyses for this revision are identified by the public revision tag. Its download guide supplies the corresponding public research materials.

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AI-use disclosure

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References

1. V. Petsiuk, A. Das, K. Saenko. RISE: Randomized input sampling for explanation of black-box models. *Proceedings of the British Machine Vision Conference (BMVC)*. 2018, https://bmva-archive.org.uk/bmvc/2018/contents/papers/1064.pdf.
2. S. Kim, J. Choi. Explaining the decisions of deep policy networks for robotic manipulations. *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. pg. 2663–2669, 2021, https://doi.org/10.1109/IROS51168.2021.9636594.
3. T. Huber, B. Limmer, E. André. Benchmarking perturbation-based saliency maps for explaining Atari agents. *Frontiers in Artificial Intelligence*. Vol. **5**, pg. 903875, 2022, https://doi.org/10.3389/frai.2022.903875.
4. H. Zhang, M. Xu, A. Dhafer, S. Yue, H. Dong, Z. D. Hao. Embodied interpretability: Linking causal understanding to generalization in vision-language-action models. *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. Vol. **306**, pg. 159526–159549, 2026, https://proceedings.mlr.press/v306/zhang26gz.html.
5. C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. C. M. Burchfiel, S. Song. Diffusion policy: Visuomotor policy learning via action diffusion. *Proceedings of Robotics: Science and Systems (RSS)*. 2023, https://doi.org/10.15607/RSS.2023.XIX.026.

6. S. Liu, L. Wu, B. Li, H. Tan, H. Chen, Z. Wang, K. Xu, H. Su, J. Zhu. RDT-1B: A diffusion foundation model for bimanual manipulation. *Proceedings of the International Conference on Learning Representations (ICLR)*. pg. 29982–30009, 2025, https://proceedings.iclr.cc/paper_files/paper/2025/hash/49f80e4d2471ad4f2edf4f5f1ab62339-Abstract-Conference.html.
7. M. Sundararajan, A. Taly, Q. Yan. Axiomatic attribution for deep networks. *Proceedings of the 34th International Conference on Machine Learning (ICML)*. Vol. **70**, pg. 3319–3328, 2017, https://proceedings.mlr.press/v70/sundararajan17a.html.
8. I. Covert, S. Lundberg, S.-I. Lee. Explaining by removing: A unified framework for model explanation. *Journal of Machine Learning Research*. Vol. **22**(209), pg. 1–90, 2021, https://jmlr.org/papers/v22/20-1316.html.
9. S. Hooker, D. Erhan, P.-J. Kindermans, B. Kim. A benchmark for interpretability methods in deep neural networks. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. **32**, pg. 9734–9745, 2019, https://papers.nips.cc/paper/9167-a-benchmark-for-interpretability-methods-in-deep-neural-networks.
10. Y. Rong, T. Leemann, V. Borisov, G. Kasneci, E. Kasneci. A consistent and efficient evaluation strategy for attribution methods. *Proceedings of the 39th International Conference on Machine Learning (ICML)*. Vol. **162**, pg. 18770–18795, 2022, https://proceedings.mlr.press/v162/rong22a.html.
11. R. Tomsett, D. Harborne, S. Chakraborty, P. Gurram, A. Preece. Sanity checks for saliency metrics. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*. Vol. **34**(4), pg. 6021–6029, 2020, https://doi.org/10.1609/aaai.v34i04.6064.
12. T. Gomez, T. Fréour, H. Mouchère. Metrics for saliency map evaluation of deep learning explanation methods. *Pattern Recognition and Artificial Intelligence (ICPRAI). Lecture Notes in Computer Science*. Vol. **13363**, pg. 84–95, 2022, https://doi.org/10.1007/978-3-031-09037-0_8.
13. A. Hedström, P. L. Bommer, K. K. Wickstrøm, W. Samek, S. Lapuschkin, M. M.-C. Höhne. The meta-evaluation problem in explainable AI: Identifying reliable estimators with MetaQuantus. *Transactions on Machine Learning Research*. 2023, https://openreview.net/forum?id=j3FK00HyfU.
14. A. Gevaert, A.-J. Rousseau, T. Becker, D. Valkenborg, T. De Bie, Y. Saeys. Evaluating feature attribution methods in the image domain. *Machine Learning*. Vol. **113**(9), pg. 6019–6064, 2024, https://doi.org/10.1007/s10994-024-06550-x.
15. I. Šimić, V. Sabol, E. Veas. Perturbation effect: A metric to counter misleading validation of feature attribution. *Proceedings of the 31st ACM International Conference on Information & Knowledge Management (CIKM)*. pg. 1798–1807, 2022, https://doi.org/10.1145/3511808.3557418.
16. P. Hase, H. Xie, M. Bansal. The out-of-distribution problem in explainability and search methods for feature importance explanations. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. **34**, pg. 3650–3666, 2021, https://proceedings.neurips.cc/paper_files/paper/2021/hash/1def1713ebf17722cbe300cfc1c88558-Abstract.html.

17. S. Blücher, J. Vielhaben, N. Strodthoff. Decoupling pixel flipping and occlusion strategy for consistent XAI benchmarks. *Transactions on Machine Learning Research*. 2024, https://openreview.net/forum?id=bIiLXdtUVM.
18. R. Hesse, S. Schaub-Meyer, S. Roth. Benchmarking the attribution quality of vision models. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. **37**, pg. 97928–97947, 2024, https://doi.org/10.52202/079017-3107.
19. X. Zheng, F. Shirani, Z. Chen, C. Lin, W. Cheng, W. Guo, D. Luo. F-Fidelity: A robust framework for faithfulness evaluation of explainable AI. *Proceedings of the International Conference on Learning Representations (ICLR)*. pg. 12772–12804, 2025, https://proceedings.iclr.cc/paper_files/paper/2025/hash/2231d1abdcc68d16c9861a58f3c25849-Abstract-Conference.html.
20. Z. Zhu, Z. Jin, J. Zhang, B. Sobieski, P. Biecek, F. Chen, J. Zhou. Faithfulness under the distribution: A new look at attribution evaluation. *Proceedings of the International Conference on Learning Representations (ICLR)*. pg. 38289–38312, 2026, https://proceedings.iclr.cc/paper_files/paper/2026/hash/409fcc9d24b549969b8b9be68b56a7be-Abstract-Conference.html.
21. S. Sridhar, A. Xue, E. Wong. Missingness bias calibration in feature attribution explanations. *Proceedings of the International Conference on Learning Representations (ICLR)*. pg. 53299–53316, 2026, https://proceedings.iclr.cc/paper_files/paper/2026/hash/57645ddd4c8532509274674ec25b76c6-Abstract-Conference.html.
22. Y. Wang, X. Wang. Benchmarking deletion metrics with the principled explanations. *Proceedings of the 41st International Conference on Machine Learning (ICML)*. Vol. **235**, pg. 51569–51595, 2024, https://proceedings.mlr.press/v235/wang24br.html.
23. N. Hama, M. Mase, A. B. Owen. Deletion and insertion tests in regression models. *Journal of Machine Learning Research*. Vol. **24**(290), pg. 1–38, 2023, https://jmlr.org/papers/v24/22-0560.html.
24. A. Mamalakis, E. A. Barnes, I. Ebert-Uphoff. Carefully choose the baseline: Lessons learned from applying XAI attribution methods for regression tasks in geoscience. *Artificial Intelligence for the Earth Systems*. Vol. **2**(1), pg. e220058, 2023, https://doi.org/10.1175/AIES-D-22-0058.1.
25. A. Atrey, K. Clary, D. Jensen. Exploratory not explanatory: Counterfactual analysis of saliency maps for deep reinforcement learning. *Proceedings of the International Conference on Learning Representations (ICLR)*. 2020, https://openreview.net/forum?id=rkl3m1BFDB.
26. Z. Cheng, X. Wu, J. Yu, W. Sun, W. Guo, X. Xing. StateMask: Explaining deep reinforcement learning through state mask. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. **36**, pg. 62457–62487, 2023, https://doi.org/10.52202/075280-2728.
27. B. Häon, K. C. Stocking, I. Chuang, C. Tomlin. Mechanistic interpretability for steering vision-language-action models. *Proceedings of the 9th Conference on Robot Learning (CoRL)*. Vol. **305**, pg. 2743–2762, 2025, https://proceedings.mlr.press/v305/haon25a.html.
28. D. D. Lundstrom, T. Huang, M. Razaviyayn. A rigorous study of integrated gradients method and extensions to internal neuron attributions. *Proceedings of the 39th International Conference on Machine Learning (ICML)*. Vol. **162**, pg. 14485–14508, 2022,

https://proceedings.mlr.press/v162/lundstrom22a.html.

29. B. Bilodeau, N. Jaques, P. W. Koh, B. Kim. Impossibility theorems for feature attribution. *Proceedings of the National Academy of Sciences*. Vol. **121**(2), pg. e2304406120, 2024, https://doi.org/10.1073/pnas.2304406120.
30. N. Akhtar, M. A. A. K. Jalwana. Towards credible visual model interpretation with path attribution. *Proceedings of the 40th International Conference on Machine Learning (ICML)*. Vol. **202**, pg. 439–457, 2023, https://proceedings.mlr.press/v202/akhtar23a.html.
31. R. Yang, B. Wang, M. Bilgic. IDGI: A framework to eliminate explanation noise from integrated gradients. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. pg. 23725–23734, 2023, https://doi.org/10.1109/CVPR52729.2023.02272.
32. L. Arras, A. Osman, W. Samek. CLEVR-XAI: A benchmark dataset for the ground truth evaluation of neural network explanations. *Information Fusion*. Vol. **81**, pg. 14–40, 2022, https://doi.org/10.1016/j.inffus.2021.11.008.
33. J. S. Kim, G. Plumb, A. Talwalkar. Sanity simulations for saliency methods. *Proceedings of the 39th International Conference on Machine Learning (ICML)*. Vol. **162**, pg. 11173–11200, 2022, https://proceedings.mlr.press/v162/kim22h.html.
34. A. Binder, L. Weber, S. Lapuschkin, G. Montavon, K.-R. Müller, W. Samek. Shortcomings of top-down randomization-based sanity checks for evaluations of deep neural network explanations. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. pg. 16143–16152, 2023, https://doi.org/10.1109/CVPR52729.2023.01549.
35. H. Jang, C. Kim, E. Yang. TIMING: Temporality-aware integrated gradients for time series explanation. *Proceedings of the 42nd International Conference on Machine Learning (ICML)*. Vol. **267**, pg. 26877–26895, 2025, https://proceedings.mlr.press/v267/jang25a.html.
36. M. Makino, Y. Asazuma, S. Sasaki, J. Suzuki. The impact of integration step on integrated gradients. *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: Student Research Workshop (EACL SRW)*. pg. 279–289, 2024, https://doi.org/10.18653/v1/2024.eacl-srw.22.
37. Robotics Diffusion Transformer team. RDT-170M model card. 2024, https://huggingface.co/robotics-diffusion-transformer/rdt-170m/blob/8aa386cac3bbfd9540676c75b3d767cc7f88a10a/README.md. (accessed 2026-11-01).
38. C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, J. Zhu. DPM-Solver++: Fast solver for guided sampling of diffusion probabilistic models. *Machine Intelligence Research*. Vol. **22**(4), pg. 730–751, 2025, https://doi.org/10.1007/s11633-025-1562-4.
39. S. Tao, F. Xiang, A. Shukla, Y. Qin, X. Hinrichsen, X. Yuan, C. Bao, X. Lin, Y. Liu, T.-K. Chan, Y. Gao, X. Li, T. Mu, N. Xiao, A. Gurha, V. N. Rajesh, Y. W. Choi, Y.-R. Chen, Z. Huang, R. Calandra, R. Chen, S. Luo, H. Su. Demonstrating GPU parallelized robot simulation and rendering for generalizable embodied AI with ManiSkill3. *Proceedings of Robotics: Science and Systems (RSS)*. 2025, https://doi.org/10.15607/RSS.2025.XXI.021.
40. X. Zhai, B. Mustafa, A. Kolesnikov, L. Beyer. Sigmoid loss for language image pre-training. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. pg.

11941–11952, 2023, https://doi.org/10.1109/ICCV51070.2023.01100.

41. C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, P. J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*. Vol. **21**(140), pg. 1–67, 2020, https://jmlr.org/papers/v21/20-074.html.
42. P. Sturmfels, S. Lundberg, S.-I. Lee. Visualizing the impact of feature attribution baselines. *Distill*. Vol. **5**(1), pg. e22, 2020, https://doi.org/10.23915/distill.00022.
43. S. Sanyal, X. Ren. Discretized integrated gradients for explaining language models. *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. pg. 10285–10299, 2021, https://doi.org/10.18653/v1/2021.emnlp-main.805.
44. L. N. Trefethen, J. A. C. Weideman. The exponentially convergent trapezoidal rule. *SIAM Review*. Vol. **56**(3), pg. 385–458, 2014, https://doi.org/10.1137/130932132.
45. M. Sotoudeh, A. V. Thakur. Computing linear restrictions of neural networks. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. **32**, pg. 14132–14143, 2019, https://proceedings.neurips.cc/paper_files/paper/2019/hash/908075ea2c025c335f4865f7db427062-Abstract.html.
46. J. Adebayo, J. Gilmer, M. Muelly, I. J. Goodfellow, M. Hardt, B. Kim. Sanity checks for saliency maps. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. **31**, pg. 9525–9536, 2018, https://papers.nips.cc/paper/8160-sanity-checks-for-saliency-maps.